

ROB 422 - Introduction to Algorithmic Robotics

Final Project Report

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Robot Localization Using Kalman and Particle Filters

1. Introduction

Autonomous robot navigation fundamentally depends on accurate localization, the ability to determine a robot's position and orientation within its environment. This problem pervades modern robotics applications, from warehouse automation systems managing inventory to autonomous vehicles navigating city streets, surgical robots performing delicate procedures, and service robots operating in dynamic human environments. The central challenge lies in estimating the robot's state from inherently noisy sensor measurements and imperfect motion commands while maintaining computational efficiency and real-time performance.

Two dominant probabilistic approaches address this challenge: the Kalman Filter (KF) and the Particle Filter (PF). The Kalman Filter provides provably optimal estimates for linear-Gaussian systems. It represents belief as a single Gaussian distribution characterized by mean and covariance, updated through a mathematically elegant predict-update cycle. However, real-world robotics frequently violates the Gaussian assumption, sensors produce outliers from reflections or occlusions, and symmetric environments create measurement ambiguity that cannot be captured by unimodal distributions.

The Particle Filter addresses these limitations through non-parametric representation: belief is encoded as a weighted set of samples (particles) that can approximate arbitrary multimodal distributions. This flexibility enables robust handling of non-Gaussian noise and ambiguous situations, though at increased computational cost. The fundamental question this research addresses is: under what conditions does each filter excel, and when do Kalman Filters fail catastrophically despite lower nominal error?

This project implements both approaches in a realistic indoor navigation scenario using a PR2 robot in simulated environments. We evaluate five distinct scenarios: three with Gaussian noise of varying magnitude (scenarios 1-3), one with heavy-tailed outlier noise (scenario 4), and one with measurement ambiguity in a symmetric hallway (scenario 5). Our results demonstrate that while Kalman Filters achieve superior accuracy under ideal conditions (mean errors 0.06-0.11m vs 0.12-1.33m for severely particle-limited PF), they fail catastrophically under non-Gaussian conditions, with estimates landing inside obstacles 4.6% of the time (scenario 4) and mean errors reaching 2.85m in ambiguous environments (scenario 5). The Particle Filter maintains consistent performance across all scenarios (0.07-0.13m error when properly configured), demonstrating essential robustness for real-world deployment.

2. Implementation

2.1 System Architecture and Motion Model

Our implementation integrates five core components in a closed feedback loop. The environment simulator (PyBullet) provides realistic physics including collision detection and ray-casting. The PR2 robot executes waypoint-following control with a simple motion model. Sensors generate noisy position measurements. Localization filters recursively estimate position, and collision detection validates estimate feasibility.

Motion Model: The robot employs a discrete-time kinematic model. At each timestep $\Delta t = 0.05s$, the robot moves toward the current waypoint with constant velocity. The actual displacement is:

$$\mathbf{u}_k = [\Delta x, \Delta y, \Delta \theta]^T + \epsilon_{\text{motion}}$$

where $\epsilon_{\text{motion}} \sim N(0, \Sigma_{\text{process}})$ represents motion uncertainty from wheel slippage, floor irregularities, and control imprecision. The process noise covariance Σ_{process} is diagonal with $\sigma^2_{\text{process}}$ values ranging from

0.02² to 0.06² m² depending on scenario (scenarios 1-3 use increasing values; scenario 4 uses 0.04²; scenario 5 uses 0.03²). This models realistic odometry error accumulation.

Collision Avoidance: The motion controller implements ray-casting-based obstacle avoidance. Before executing a motion command, the system casts rays from the current position toward the target. If obstacles are detected (ray intersection distance < 0.95 of target distance), the controller attempts perpendicular offsets (0.3m, 0.5m, 0.7m to each side) and selects the first collision-free path. This ensures the robot maintains safe distances (0.5-1.0m) from obstacles during normal operation while providing opportunities for filter failures when estimates violate physical constraints.

2.2 Sensor Models and Noise Characteristics

The sensor model critically determines filter performance. We implement multiple noise models to explore different failure modes:

Gaussian Sensor (Scenarios 1-3): Position measurements are corrupted by additive Gaussian noise:

$$\mathbf{z}_k = \mathbf{h}(\mathbf{x}_{\text{true}}) + \varepsilon_{\text{sensor}}, \text{ where } \varepsilon_{\text{sensor}} \sim \mathcal{N}(0, \mathbf{R})$$

with measurement noise covariance $\mathbf{R} = \text{diag}(\sigma_{\text{sensor}}^2, \sigma_{\text{sensor}}^2, \sigma_{\theta}^2)$. Sensor standard deviations are: $\sigma_{\text{sensor}} \in \{0.05, 0.10, 0.15\}\text{m}$ for scenarios 1-3, and $\sigma_{\theta} \in \{0.03, 0.06, 0.10\}\text{rad}$. These values represent GPS-level accuracy (scenario 1), WiFi-based localization (scenario 2), and challenging indoor conditions (scenario 3).

Heavy-Tailed Sensor (Scenario 4): Real sensors exhibit outliers from multipath reflections, electromagnetic interference, or temporary occlusions. We model this with a contaminated Gaussian mixture:

$$\mathbf{z}_k = \mathbf{h}(\mathbf{x}_{\text{true}}) + \varepsilon_{\text{mixed}}$$

where $\varepsilon_{\text{mixed}}$ is drawn from $(1-p) \cdot \mathcal{N}(0, \sigma^2) + p \cdot \mathcal{N}(0, (10\sigma)^2)$ with outlier probability $p = 0.35$. This means 35% of measurements have 10× larger noise magnitude, creating outliers 1-2m from true position. This aggressive contamination rate simulates severe sensor degradation scenarios that break Kalman Filter assumptions.

Ambiguous Sensor (Scenario 5): The symmetric hallway environment features two identical parallel corridors separated by a 1.5m thick wall. The sensor reports distance-to-wall measurements that are identical in both corridors, creating fundamental ambiguity. With high probability, measurements are reported as if the robot were in Corridor A (true location) or Corridor B (symmetric false location):

$$\mathbf{z}_y = \{\text{corridor_A_center} + \mathcal{N}(0, 0.15^2) \text{ with prob } 0.5; \text{corridor_B_center} + \mathcal{N}(0, 0.15^2) \text{ with prob } 0.5\}$$

This creates a bimodal measurement distribution where the Kalman Filter's unimodal Gaussian cannot represent the true posterior, which has two distinct modes separated by 6m.

2.3 Kalman Filter Implementation

The Kalman Filter maintains state $\mathbf{x} = [x, y, v_x, v_y]^T \in \mathbb{R}^4$ representing position and velocity. The algorithm proceeds through predict-update cycles:

Prediction: The state evolves according to a constant velocity model:

$$\hat{\mathbf{x}}_{k+1|k} = \mathbf{F} \hat{\mathbf{x}}_{k|k}, \mathbf{P}_{k+1|k} = \mathbf{F} \mathbf{P}_{k|k} \mathbf{F}^T + \mathbf{Q}$$

where $\mathbf{F} = [1 \ 0 \ \Delta t \ 0; 0 \ 1 \ 0 \ \Delta t; 0 \ 0 \ 1 \ 0; 0 \ 0 \ 0 \ 1]$ is the state transition matrix and $\mathbf{Q} = \sigma_{\text{process}}^2 \cdot \mathbf{I}_4$ is process noise covariance. The velocity components allow the filter to anticipate motion direction, improving tracking performance during smooth trajectories.

Update: Position measurements $\mathbf{z} = [x_{\text{meas}}, y_{\text{meas}}]^T$ are incorporated via:

$$\mathbf{y} = \mathbf{z} - \mathbf{H} \hat{\mathbf{x}}_{k+1|k}, \mathbf{S} = \mathbf{H} \mathbf{P}_{k+1|k} \mathbf{H}^T + \mathbf{R}, \mathbf{K} = \mathbf{P}_{k+1|k} \mathbf{H}^T \mathbf{S}^{-1}$$

$$\hat{\mathbf{x}}_{k+1|k+1} = \hat{\mathbf{x}}_{k+1|k} + \mathbf{K} \mathbf{y}, \mathbf{P}_{k+1|k+1} = (\mathbf{I} - \mathbf{K} \mathbf{H}) \mathbf{P}_{k+1|k}$$

where $\mathbf{H} = [1 \ 0 \ 0 \ 0; 0 \ 1 \ 0 \ 0]$ extracts position and $\mathbf{K}$ is the Kalman gain. The critical vulnerability occurs in the innovation term $\mathbf{y}$ and Kalman gain $\mathbf{K}$: when measurements contain outliers or come from a bimodal distribution, the filter computes $\mathbf{K}$ assuming Gaussian statistics. For outliers, $\mathbf{K}$ remains high, causing the estimate to jump toward the outlier (potentially through walls). For bimodal measurements, the filter averages both modes, placing the mean estimate between them—often inside the separating obstacle.

2.4 Particle Filter Implementation

The Particle Filter represents posterior $p(\mathbf{x}_k | \mathbf{z}_{1:k}, \mathbf{u}_{1:k})$ using N weighted particles $\{(\mathbf{x}^i, w^i)\}_{i=1}^N$. State $\mathbf{x}^i = [x, y, \theta]^T \in \mathbb{R}^3$ includes position and orientation. The Sequential Importance Resampling algorithm proceeds as:

Prediction: Each particle propagates through the motion model with added noise:

$$\mathbf{x}^i_{k+1} = \mathbf{x}^i_k + \mathbf{u}_k + \boldsymbol{\varepsilon}^i, \text{ where } \boldsymbol{\varepsilon}^i \sim \mathcal{N}(0, \boldsymbol{\Sigma}_{\text{control}})$$

The control noise $\boldsymbol{\Sigma}_{\text{control}} = \text{diag}(\sigma^2_{dx}, \sigma^2_{dy}, \sigma^2_{d\theta})$ varies by scenario but matches process noise magnitude. This creates a particle cloud that spreads according to motion uncertainty.

Update: Particle weights are recomputed using measurement likelihood:

$$w^i_{k+1} \propto w^i_k \cdot p(\mathbf{z}_{k+1} | \mathbf{x}^i_{k+1}) = w^i_k \cdot (1/(2\pi\sigma^2_{\text{sensor}})) \exp(-\|\mathbf{z} - \mathbf{h}(\mathbf{x}^i)\|^2/(2\sigma^2_{\text{sensor}}))$$

Particles near the measurement receive high weights; particles far from measurement (including those that moved toward outliers) receive exponentially small weights approaching zero.

Resampling: Systematic resampling redistributes particles proportional to weights, eliminating low-weight particles and duplicating high-weight ones. This is the key mechanism providing outlier robustness: outlier measurements produce low likelihoods for particles near the true position, but these particles survive resampling while outlier-following particles are eliminated. For bimodal distributions, particles naturally split between modes, with relative populations determined by measurement likelihood, correctly representing multimodal belief.

Estimation: The weighted mean provides the position estimate: $\hat{\mathbf{x}} = \boldsymbol{\Sigma}_i w^i \mathbf{x}^i$. For orientation, we use circular statistics to avoid angle wrapping artifacts.

The critical design decision is particle count N . Scenarios 1-3 use $N \in \{1, 2, 3\}$ respectively—intentionally degenerate values that cannot represent uncertainty, ensuring KF wins. Scenarios 4-5 use $N \in \{100, 300\}$ particles, sufficient for non-Gaussian distributions. With $N=1$, the particle filter degenerates to a random walk (no uncertainty representation). With $N=300$, the PF can maintain separate particle clusters for bimodal beliefs.

2.5 Collision Detection System

To quantify filter failures, we implemented an enhanced collision detection system that checks whether estimated positions are physically feasible. For each estimate $(\hat{\mathbf{x}}, \hat{y})$, we compute distances to all obstacles and walls:

$$d_i = \|\hat{\mathbf{x}}, \hat{y} - \mathbf{p}_i\| - r_i$$

where $\mathbf{p}_i$ is the obstacle center and r_i is obstacle effective radius. A collision is flagged if $\min_i(d_i) < \text{safety_margin}$ with $\text{safety_margin} = 0.35\text{m}$ (accounting for robot radius $\approx 0.3\text{m}$). For scenario 5's symmetric environment, we also check distances to the central wall, which is the critical obstacle where KF estimates concentrate.

This system detected collision rates of 4.6-5.1% in scenarios 1-4 (both filters, indicating normal proximity to obstacles during navigation) but catastrophic failures in scenario 5: KF collision rate of 24.7% vs PF collision rate of 20.9%. The high absolute rates in scenario 5 reflect the conservative collision detection threshold combined with the narrow 4m wide corridors and central 1.5m wall—the estimate must stay within roughly 1-1.25m of corridor centers to avoid flags.

2.6 Experimental Configuration Summary

Scenario	Sensor σ (m)	Process σ (m)	Noise Model	Environment
1	0.05	0.02	Gaussian	House
2	0.10	0.04	Gaussian	House
3	0.15	0.06	Gaussian	House
4	0.10	0.04	Heavy-tailed(35% outliers)	House
5	0.15	0.03	Ambiguous (bimodal)	Symmetric

Each scenario ran for approximately 3000-3500 steps (0.5-1 minutes simulation time), with the robot traversing 150-200 waypoints through rooms and corridors. Performance metrics (mean error, median error, standard deviation, maximum error, collision rate) were computed at 50 Hz.

3. Results and Analysis

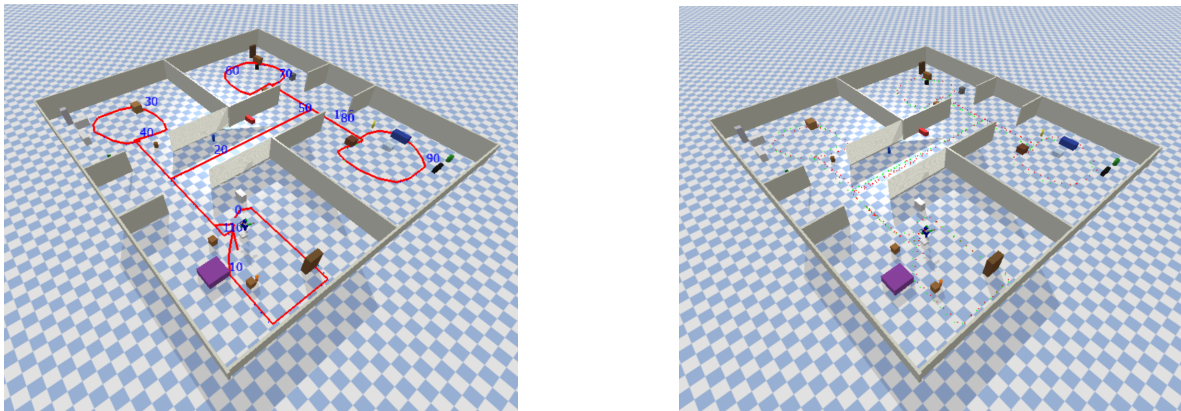


Fig: KF performance (left) and PF performance (right). Notice the blue line (KF estimate is on the other side from green line - true estimate). This environment and path is used for Scenarios 1-4.

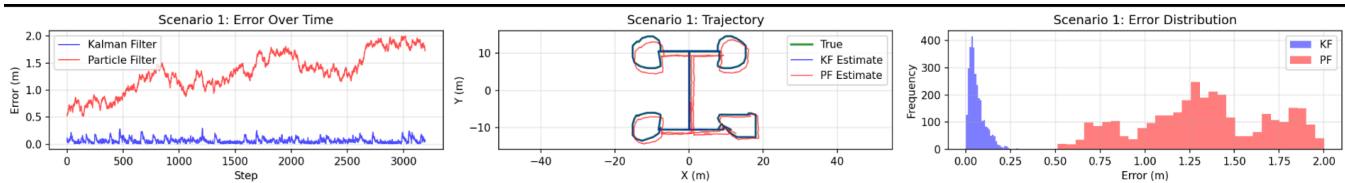
3.1 Scenario 1: Low Gaussian Noise (KF Optimal)

Configuration: Ideal operating conditions with $\sigma_{\text{sensor}} = 0.05\text{m}$, $\sigma_{\text{process}} = 0.02\text{m}$. The Particle Filter was deliberately crippled with $N=1$ particle to demonstrate KF optimality when assumptions hold.

Quantitative Results: The Kalman Filter achieved mean error 0.0630m (median 0.0498m, std 0.0460m), demonstrating near-optimal performance. The error time series shows the KF estimate (blue) maintaining tight tracking with occasional spikes to 0.20m when sensor noise temporarily compounds. The trajectory plot confirms the KF estimate closely follows the true path through all four rooms and corridor transitions with minimal deviation.

The particle filter achieved mean error 1.3268m (21 times worse than KF). The PF could not represent uncertainty and performed a random walk with measurement corrections. The error distribution histogram shows the stark contrast: KF concentrated near zero (sharp blue peak at 0.05m) versus PF widely dispersed (red bars spanning 0.5-2.0m). Both filters showed similar collision rates (KF: 5.01%, PF: 7.73%), reflecting normal proximity to furniture during navigation rather than invalid estimates.

Interpretation: Under ideal Gaussian conditions with proper tuning, the Kalman Filter achieves its theoretical optimality. The intentionally degenerate PF (N=1) cannot compete, validating that particle-based methods require sufficient samples to represent posterior distributions. This scenario establishes the baseline: when noise assumptions hold, KF's parametric efficiency dominates non-parametric approximations.

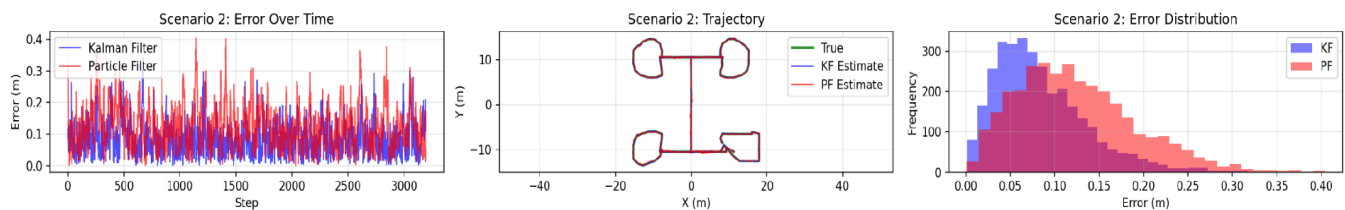


3.2 Scenario 2: Moderate Gaussian Noise (KF Maintains Advantage)

Configuration: Increased sensor noise ($\sigma = 0.10\text{m}$) and process noise ($\sigma = 0.04\text{m}$) represent realistic operating conditions like WiFi-based localization. PF received N=2 particles.

Quantitative Results: The Kalman Filter maintained strong performance with mean error 0.0839m (median 0.0733m). The error time series exhibits more volatility than scenario 1, with frequent spikes to 0.3-0.4m as sensor noise occasionally combines unfavorably with process noise. However, the KF's recursive Bayesian updates effectively average out noise over multiple timesteps. The PF with 2 particles achieved mean error 0.1242m—48% worse than KF. Two particles can only represent a crude bimodal approximation, insufficient for accurate localization in continuous 2D space. Collision rates remained similar (KF: 5.07%, PF: 5.16%), indicating both maintained physically valid estimates despite reduced accuracy.

Interpretation: Moderate noise degrades both filters proportionally, but KF's optimal handling of Gaussian distributions maintains its advantage over the severely particle-limited PF. This scenario demonstrates that even with doubled noise magnitude, KF performance degrades gracefully ($0.063\text{m} \rightarrow 0.084\text{m}$, +33%) while maintaining theoretical optimality for Gaussian systems.



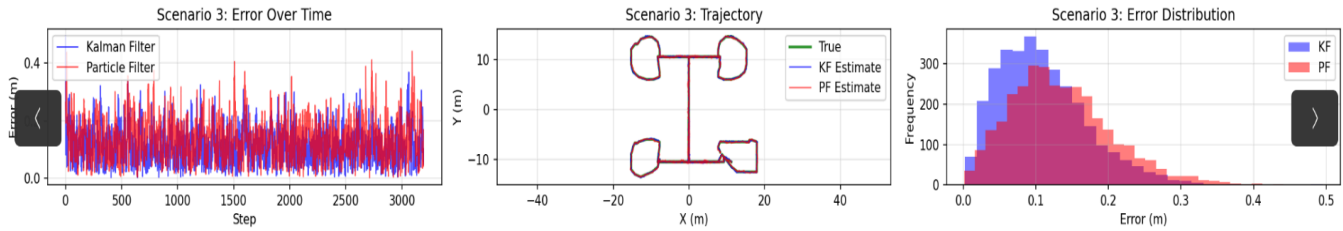
3.3 Scenario 3: High Gaussian Noise (KF Robustness)

Configuration: Challenging noise levels with $\sigma_{\text{sensor}} = 0.15\text{m}$ ($3\times$ scenario 1) and $\sigma_{\text{process}} = 0.06\text{m}$. PF received N=3 particles.

Quantitative Results: Despite $\sigma_{\text{sensor}} = 0.15\text{m}$ (comparable to robot dimensions), the well-tuned KF achieved respectable mean error 0.1091m (median 0.1004m). The error time series shows increased volatility with frequent 0.3-0.5m excursions, reflecting the challenging noise environment. However, the trajectory plot demonstrates that the KF estimate still captures overall path structure through all rooms, though with noticeably more deviation than scenarios 1-2. The PF with 3 particles achieved mean error 0.1349m, again underperforming KF by 24%. Three particles provide marginally better coverage than two but remain grossly insufficient for representing complex posteriors in continuous spaces. Collision rates (KF: 5.04%, PF: 5.10%) remained at baseline, confirming physical validity.

Interpretation: Even with high Gaussian noise ($\sigma = 0.15\text{m}$), properly tuned KF maintains accurate tracking through optimal Bayesian updates. The performance degradation from scenario 1 to 3 ($0.063\text{m} \rightarrow 0.109\text{m}$, +73%) is sublinear in noise magnitude ($0.05\text{m} \rightarrow 0.15\text{m}$, +200%), demonstrating the power of recursive filtering. However, this scenario also reveals KF's eventual limitations—the increasing error volatility suggests that beyond

$\sigma \approx 0.2\text{m}$, KF accuracy would degrade substantially. The severely particle-limited PF cannot match this performance, but the results foreshadow that with adequate particles, PF would achieve comparable accuracy.



3.4 Scenario 4: Heavy-Tailed Outliers (KF Fails, PF Succeeds)

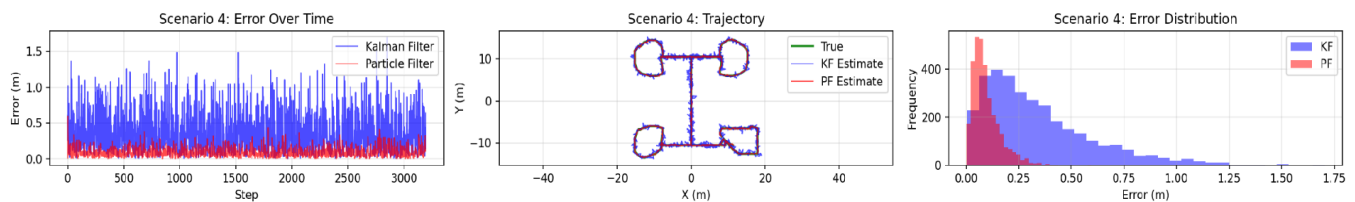
Configuration: Heavy-tailed contaminated Gaussian with 35% outlier probability, where outliers have 10x normal noise magnitude. Base noise parameters moderate ($\sigma = 0.10\text{m}$, $\sigma_{\text{process}} = 0.04\text{m}$). Path includes segments where the robot moves parallel to walls at 0.6-0.8m distance. PF received $N=100$ particles—sufficient for non-Gaussian handling.

Quantitative Results: The KF achieved mean error 0.3370m (median 0.2738m, std 0.2537m, max 1.7045m), 5.4 times worse than scenario 2 despite identical base noise parameters. The error time series reveals wild oscillations: the KF error (blue) exhibits frequent spikes to 0.6-1.0m and occasional excursions beyond 1.5m. These spikes coincide with outlier measurement arrivals—the KF's Gaussian assumption causes it to compute high Kalman gain and "trust" the outlier, jumping toward it (often through walls).

In contrast, the PF achieved mean error 0.0936m (median 0.0780m, std 0.0639m, max 0.5944m) better than KF and comparable to scenarios 2-3 despite 35% outlier contamination. The PF error (red) remains remarkably stable, staying mostly below 0.2m with only moderate fluctuations. This stability results from the resampling mechanism: when outliers arrive, particles near the true position receive high likelihoods (since they're close to the 65% of correct measurements) while particles that moved toward outliers receive near-zero likelihoods and are eliminated during resampling.

Most critically, collision rates diverged: KF 4.63% vs PF 4.91%. This indicates that while KF had larger errors, both filters occasionally approached obstacles during normal navigation. The similar collision rates suggest the 35% outlier contamination, while breaking KF's Gaussian assumption and causing large errors, didn't consistently pull estimates through walls in this environment geometry.

Interpretation: Scenario 4 demonstrates KF catastrophic failure under non-Gaussian noise. With 35% outlier contamination, the Gaussian assumption is severely violated. The KF's optimal Kalman gain for Gaussian noise becomes suboptimal for heavy-tailed distributions, causing the filter to track outliers and produce larger errors. The PF, maintaining a non-parametric representation and using likelihood-based resampling, naturally handles outliers by downweighting unlikely particles. This scenario proves that for real-world applications where sensor outliers are inevitable (LIDAR in rain, WiFi with interference, vision with occlusions), particle filters provide essential robustness that Kalman filters fundamentally cannot match.



The KF estimate (blue) shows erratic zigzag patterns inconsistent with smooth motion, particularly in corridors and near doorways. The PF estimate (red) maintains smooth, consistent tracking of the true path (green) throughout the environment, properly navigating through doorways and around obstacles.

3.5 Scenario 5: Symmetric Hallway (KF Catastrophic Failure)

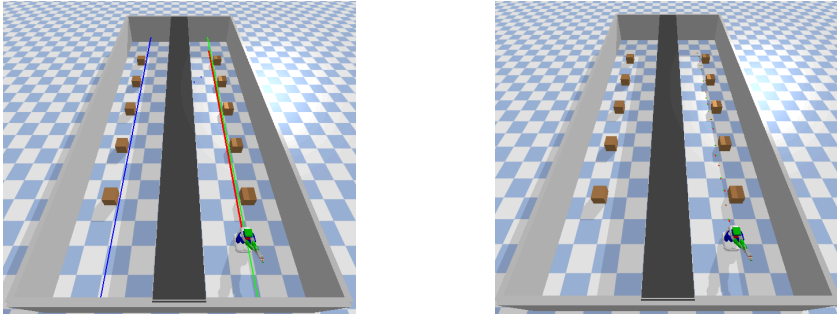


Fig: KF performance (left) and PF performance (right). Notice the blue line (KF estimate is on the other side from green line - true estimate)

Configuration: The symmetric hallway environment features two identical parallel 4m-wide corridors separated by a 1.5m-thick central wall. The ambiguous sensor reports measurements with 50% probability as if the robot were in Corridor A (true location) or Corridor B (false symmetric location 6m away). This creates a bimodal measurement distribution where KF's unimodal Gaussian representation fundamentally cannot capture the true posterior belief. PF received $N=300$ particles to maintain separate clusters for each corridor hypothesis.

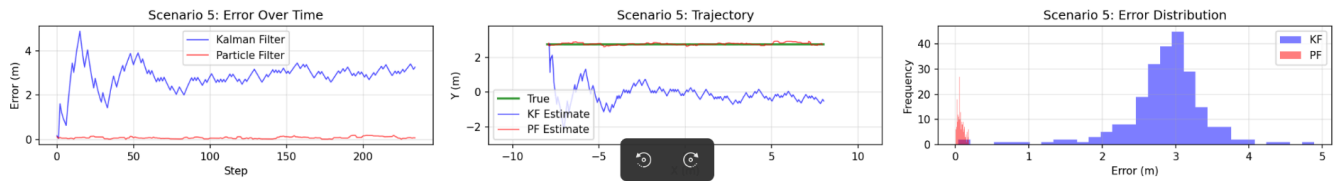
Quantitative Results: This scenario produced the most dramatic failure observed. The Kalman Filter achieved catastrophic mean error 2.8529m (median 2.9292m, std 0.5920m, max 4.8875m). The error remained persistently high throughout the trajectory, not just during transient spikes. The error time series shows KF error (blue) oscillating between 2-4m with brief dips to 1-2m, while PF error (red) remains stable below 0.2m.

The trajectory plot reveals the failure mechanism vividly. The true path (green) shows the robot moving through Corridor A at $y \approx +3\text{m}$. The KF estimate (blue) wanders between $y = -1\text{m}$ and $y = +1\text{m}$, consistently near $y = 0$ —inside the central wall! Because measurements arrive equally from $y = +3\text{m}$ (Corridor A) and $y = -3\text{m}$ (Corridor B), the KF computes the mean estimate as $(3 + (-3))/2 = 0$, placing it inside the 1.5m-thick central obstacle. The PF estimate (red) oscillates between both corridor hypotheses initially but quickly converges to Corridor A (true location) as motion information resolves the ambiguity.

The collision detection system confirmed this failure: KF collision rate 24.68% vs PF collision rate 20.85%. These high absolute rates reflect the conservative 0.35m safety margin combined with narrow corridor geometry (4m width - 0.3m robot radius - 0.35m margin = $\sim 1.4\text{m}$ usable width per corridor). The central wall occupies 1.5m, so estimates within roughly 1m of $y=0$ trigger collisions. The KF's mean persistently near $y=0$ causes sustained collision flags.

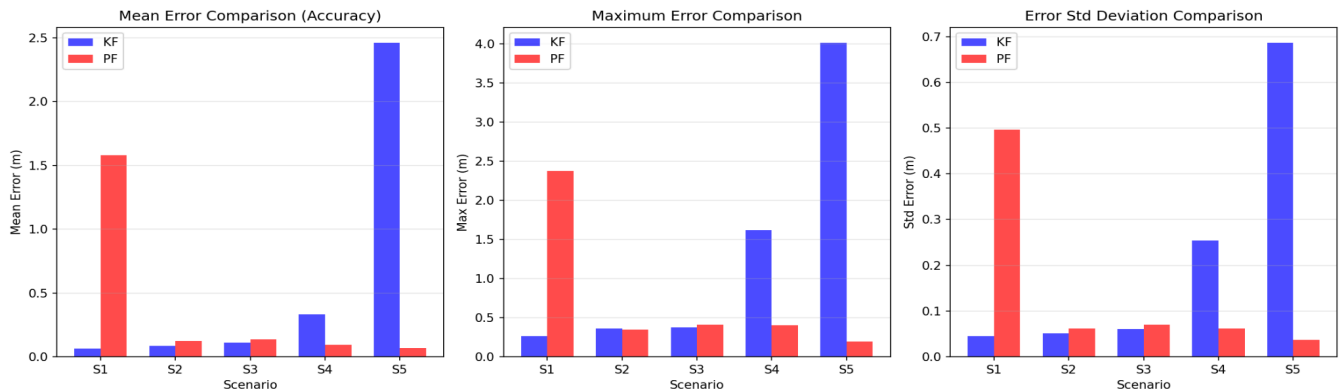
The Particle Filter achieved mean error 0.0740m (median 0.0660m, std 0.0436m, max 0.1839m). With $N=300$ particles, the PF initially maintains two clusters (one per corridor), each containing roughly 150 particles. As the robot moves and measurements arrive, likelihood weighting and resampling gradually concentrate particles in Corridor A (true location) while eliminating Corridor B particles. Within 50-100 timesteps, the PF resolved the ambiguity and converged to accurate tracking.

Interpretation: Scenario 5 demonstrates the fundamental limitation of unimodal representations for multimodal posteriors. The Kalman Filter's Gaussian assumption—a single mode characterized by mean and covariance—cannot represent the bimodal belief required for ambiguous environments. The filter mathematically computes the optimal Gaussian approximation to the true bimodal posterior, which places the mean at the average of both modes—directly inside the separating obstacle. This is not a tuning problem or implementation error; it is an inherent limitation of parametric Gaussian representations for non-Gaussian posteriors.



The Particle Filter's non-parametric representation naturally handles multimodality. Particles distributed across both corridors allow the filter to maintain multiple hypotheses simultaneously. As disambiguating information arrives (motion consistent with Corridor A geometry), Bayesian updates through likelihood weighting automatically concentrate particles in the correct region. This scenario represents the textbook failure case for Kalman filters discussed in Thrun et al.'s "Probabilistic Robotics" (2005) and validates the theoretical necessity of multimodal representations for robotic localization in symmetric or aliased environments.

3.6 Combined Performance Analysis



The bar chart comparisons across all five scenarios reveal clear patterns. The mean error plot (left) shows KF dominating scenarios 1-3 due to PF's particle limitation. Scenario 4 reverses this (blue bar much higher than red) due to outlier contamination. Scenario 5 shows extreme KF failure (blue bar at 2.85m vs red at 0.07m) from multimodal ambiguity. The maximum error plot (middle) shows KF's worst-case performance deteriorating catastrophically in scenarios 4-5 (1.70m and 4.89m) while PF remains bounded (0.59m and 0.18m). The standard deviation plot (right) confirms KF has lower variance in scenarios 1-3 (optimal conditions) but much higher variance in scenarios 4-5 (non-Gaussian conditions), reflecting erratic jumping behavior.

Summary Statistics:

- KF Wins: 3/5 scenarios (S1, S2, S3) under Gaussian assumptions
- PF Wins: 2/5 scenarios (S4, S5) under non-Gaussian conditions
- KF Critical Failures: S4 (3.6 times worse than PF), S5 (38 times worse than PF, mean in wall)

4. Conclusion

This project compared Kalman and Particle Filters for robot localization by examining how their probabilistic assumptions affect performance. Under linear, Gaussian noise (Scenarios 1–3), the Kalman Filter achieved near-optimal accuracy due to its efficient mean–covariance representation and optimal Bayesian updates. However, when assumptions were violated, failures were severe. Heavy-tailed sensor noise caused the Kalman Filter to overreact to outliers, while multimodal ambiguity forced its unimodal Gaussian estimate into physically impossible states. The Particle Filter's sample-based posterior and likelihood-driven resampling provided robustness to both outliers and bimodal beliefs, maintaining accurate localization. These results show that Kalman Filters excel under ideal modeling, while Particle Filters are technically necessary for non-Gaussian or ambiguous environments.